MACHINE LEARNING FOR HEART ATTACK SCREENING: CAN WE IDENTIFY RISK BEFORE A CRISIS OCCURS?

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A. INTRODUCTION

1. WHY HEART ATTACK SCREENING MATTERS?

- **Annual Occurrence**: Approximately 805,000 Americans have a heart attack (605,000 first, 200,000 recurrent). (CDC,2023)
- **Costly**: According to the **American Heart Association**, the average cost of a heart attack in the United States is estimated to be over **$20,000**. (Resolve Medical Bills, 2023)
- Traditional screening may miss complex risk interactions. Machine learning enables personalized, data-driven early risk detection.

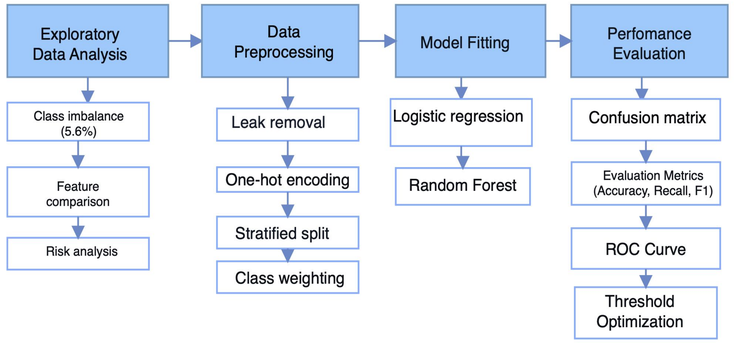
2. DATASET AND CHALLENGES

Dataset: self-survey from **237,630** U.S. adults (Kaggle).
Features: Demographics (age,sex,race), body metric(BMI, weight, height), clinical conditions (diabetes,arthritis,mobility limitations), behavioral risks(smoking, alcohol), self-reported health status.
Challenge 1: **Self-reported survey data** introduces potential bias and limits direct clinical generalization.
Challenge 2: Class imbalance: Heart attack cases account for only 5.6% of the population - the model must handle class imbalance while maintaining clinical interpretability.

3. GOALS

- Build a **heart attack screening prototype** that can identify high-risk individuals in a low-prevalence population (5.6%).
- Develop an interpretable ML-based screening model for early heart attack risk detection.
- Handle severe class imbalance (5.6% prevalence) while prioritizing sensitivity over accuracy.
- Provide actionable insights to support preventive healthcare decision-making.

B. METHODOLOGY



Data Imbalance: Individuals with a history of heart attack represent 5.6% of the dataset. This was addressed using stratified sampling, class-weighted models, and threshold optimization to prioritize sensitivity.

C. INSIGHTS

HEART ATTACK DOESN'T ONLY HIT "HIGH-RISK" PATIENTS, IT OFTEN HITS THE UNPREPARED.SMALL RISK TODAY CAN BECOME A HEART ATTACK TOMORROW.

Poor general health, kidney disease, COPD, and mobility limitations significantly increase heart attack odds (OR > 1). Chest-related medical history shows the strongest association (~2x). These findings reinforce comorbidity burden as a critical screening signal.

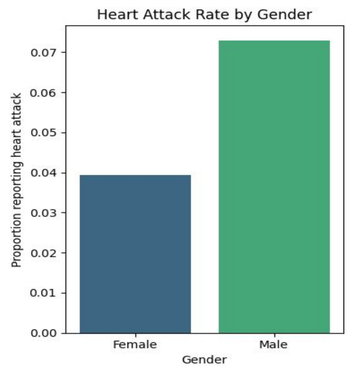


Figure 1. Heart Attack Rate by Gender

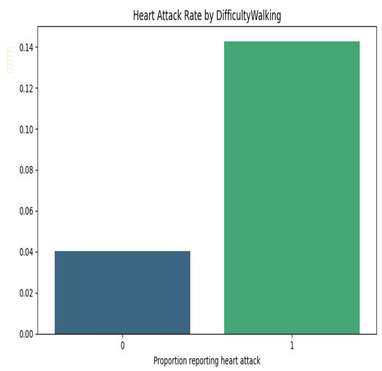


Figure 2.. Heart Attack Rate by Difficult Walking

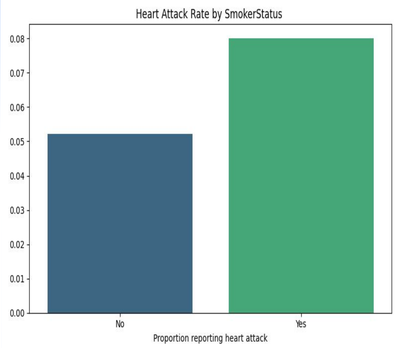


Figure 3.. Heart Attack Rate by Smoker status

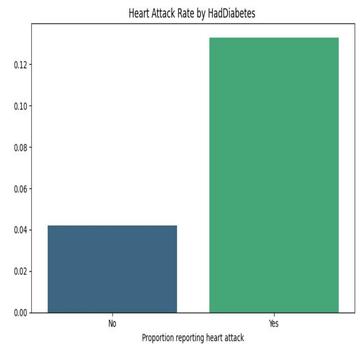


Figure 4. Heart Attack Rate by Had Diabetes

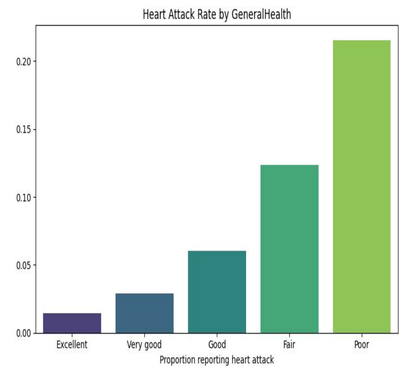


Figure 5.. Heart Attack Rate by General Health

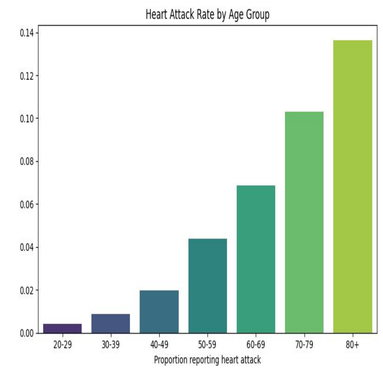
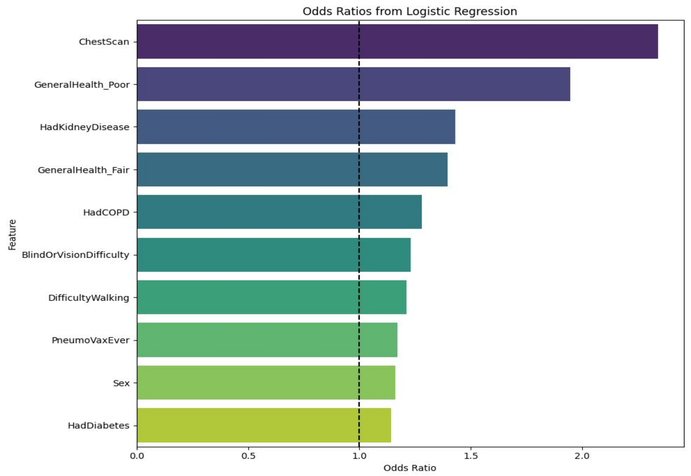


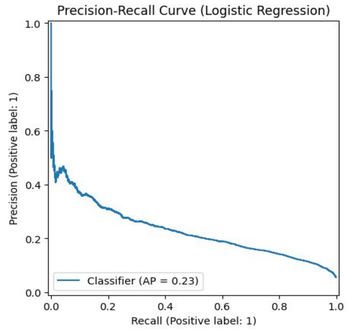
Figure 6.. Heart Attack Rate by Age Group

D. FINAL MODEL

Logistic Regression: Baseline screening model; **high interpretability** (direction of risk factors) and strong discrimination (**ROC-AUC ≈ 0.83**).
Random Forest: Non-linear benchmark; provides feature importance, but showed **low recall** under imbalance, making it less suitable for screening.
Final MODEL : LOGISTIC REGRESSION was selected because screening requires **high sensitivity**; threshold tuning improved recall (**0.775 TO 0.888 at threshold = 0.35**).



- ChestScan (~2.3x) – individuals reporting chest scan history have more than **2x higher odds** of heart attack
-Poor General Health (~2.0x) – strong association with increased risk.
-Kidney Disease (~1.5x) – significant comorbidity risk factor.



E. DISCUSSION

Logistic Regression achieved strong discrimination and high sensitivity for screening.

Class imbalance and survey-based design limit clinical generalizability.
Future validation on clinical datasets is needed.

References:
Centers for Disease Control and Prevention. (2023). *Heart attack facts*. U.S. Department of Health & Human Services. https://www.cdc.gov/heartdisease/heart_attack.htm
Resolve Medical Bills. (2023, October 20). *The true cost of a heart attack in the US*. Resolve Medical Bills. https://www.resolvemedicalbills.com/blog/the-true-cost-of-a-heart-attack-in-the-us